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Machine learning-based prediction of household sanitation facility access in Sub-Saharan Africa: insights from DHS data (2012–2024)

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Abstract

Background Access to adequate sanitation facilities remains a major global public health challenge, with an estimated 3.6 billion people lacking safely managed sanitation. Most of these populations reside in low- and middle-income countries, particularly in sub-Saharan Africa (SSA). Sanitation utilization is influenced by socioeconomic, demographic, housing, and environmental factors, contributing to the burden of fecal–orally transmitted diseases. This study aimed to predict household sanitation facility access in SSA using machine learning techniques.

Methods Data were obtained from Demographic and Health Surveys (DHS) conducted between 2012 and 2024 across 34 SSA countries, comprising 500,845 households. A stratified two-stage cluster sampling design ensured national representativeness. Five supervised machine learning models, Random Forest (RF), Decision Tree (DT), Extreme Gradient Boosting (XGBoost), Logistic Regression (LR), and Artificial Neural Network (ANN) were developed. The dataset was split into training (80%) and testing (20%) subsets. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied, followed by 10-fold cross-validation. Model performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. Feature importance was interpreted using Shapley Additive Explanations (SHAP).

Results The prevalence of unimproved sanitation facility access was 49.85%, with substantial disparities observed within and across countries. The Random Forest model demonstrated the best predictive performance (accuracy = 80.61%, F1-score = 0.8377), outperforming other algorithms. SHAP analysis identified shared toilet (feature importance = 0.233), education level (0.204), and wealth index (0.094) as the most influential predictors, followed by residence, electricity access, age of the household head, drinking water source, household water treatment, and cooking fuel type. Media access, sex of the household head, handwashing facility, water access, soap presence, location of water source, marital status, and household size also contributed notably. The number of under-five children had minimal predictive influence.

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Conclusion Machine learning models, particularly Random Forest, effectively predicted sanitation facility access in sub-Saharan Africa, highlighting key socioeconomic and infrastructural determinants. Targeted interventions such as economic empowerment, health education, infrastructure investment, and media engagement are essential to reduce reliance on unimproved sanitation and accelerate progress toward the Sustainable Development Goals.

Keywords Households, Machine learning, Sanitation facility, Sub-Saharan africa, Toilet

Text box 1. Contributions to the literature

- This study pioneers the use of multi-country machine learning models to predict household sanitation access across sub-Saharan Africa using harmonized DHS data (2012–2024).
- It demonstrates the utility of interpretable artificial intelligence (SHAP analysis) for identifying socioeconomic and infrastructural drivers of sanitation disparities.
- The findings advance evidence-based targeting of WASH interventions by integrating predictive analytics into public health decision-making.
- It provides a scalable framework for applying data-driven methods to monitor progress toward SDG 6.2 in resource-limited settings.

Introduction

Improving public access to sanitation services in a rapidly urbanizing world is an increasingly important, yet challenging issue for governments, especially in less developed nations. The Joint Monitoring Program (JMP) of the World Health Organization (WHO) defines improved sanitation facilities as a sanitation system that disposes of excreta in a way that lowers the risk of feco-oral transmission to users and the environment [1]. These facilities include composting toilets, ventilated improved pit latrines, septic tanks or pit latrines, flush or pour-flush to a piped sewer system, and pit latrines with slabs [2, 3]. Every year, 775,000 people around the world die as a result of poor sanitation [4]. In 2015, access to better sanitation was still unavailable to 2.4 billion people globally, the majority living in Sub-Saharan Africa (SSA) [2]. According to WHO and UNICEF reports in 2020, globally, about 3.6 billion people lack safely managed sanitation services, and more than 673 million practice open field defecation [5]. According to WHO's 2022 statistics, only 57% of the global population, roughly 4.6 billion people, had access to safely managed sanitation services, while the remainder relied on unimproved sanitation facilities [6].

In SSA, 700 million people, or about 63.6% of the population, lacked access to improved sanitation in 2015 [1]. While the region's population has nearly doubled from 1990 To 2015, access to improved sanitation has increased by only 6% points within the same period [7]. Like other least-developed regions, SSA has limited coverage and did not achieve Target 10 of the Millennium Development Goals (MDGs), but it did make progress [8]. Sustainable Development Goals (SDG) aimed to eliminate open defecation and guarantee that everyone has access to fair and sufficient sanitation by 2030 [9]. In

addition to being included in the SDGs, several low- and middle-income nations that have put in place extensive sanitation programs have acknowledged sanitation as a development and public health priority [10]. In 2020, only one-fifth of the population had access to safely managed sanitation facilities [6].

Poor sanitation and a lack of adequate sanitation facilities have a detrimental effect on the health of society, the economy, and the environment [11, 12]. It contributes to the spread of diseases like cholera, diarrhea, dysentery, typhoid, Polio, and hepatitis A, contaminates drinking water sources, degrades the environment, and causes malnutrition and low school attendance in children [13–16]. Globally, 88% of diarrheal illnesses are caused by inadequate sanitation, contaminated drinking water, and poor hygiene [17], and more than 297,000 children under the age of 5 died in 2019 for the same reason [18]. Economically, it leads to significant healthcare costs and productivity losses, hindering labor productivity and economic growth due to illness-related absenteeism and healthcare expenses [19]. The estimated global economic loss in 2015 for diseases associated with poor sanitation and lost productivity was 222.9 billion dollars [20].

Different studies identified different factors, such as sociodemographic factors, housing, and environmental factors, that will influence the household's access and utilization. Age and sex of head of the households, educational attainment, wealth index, having under-five children in the household, household size, access to electricity, water source, access to water source, number of rooms, housing condition, and place of residence had a significant association with households' sanitation facility preference [21–26]. The United Nations formally recognized access to safe sanitation as a human right in 2012 and the adoption of the Sustainable Development Agenda (SDA) in 2015, where Goal 6.2 called for universal access to adequate and equitable sanitation and hygiene by 2030 to prevent human interactions with fecal pathogens to reduce incidences of diseases [27], further demonstrate the significance of sanitation in enhancing citizens' well-being and advancing sustainable development. Despite improvements, it is unlikely that the SDGs' sanitation target will be met [28].

Despite extensive evidence linking socioeconomic and demographic factors to sanitation access, most studies have relied on traditional regression models within single-country contexts. This study advances the field

by applying interpretable machine learning techniques to predict household sanitation facility access across 34 Sub-Saharan African countries (2012–2024). Machine learning tools are recognized to increase the power of near-term prediction and to generate optimal conditions for sanitation management development [29, 30]. Using SHAP analysis, we quantify and interpret the relative influence of determinants, capturing complex, nonlinear relationships often missed by conventional methods. By integrating data-driven prediction with public health planning, the study provides actionable insights for reducing sanitation inequities, optimizing resource allocation, and strengthening climate-resilient WASH programs. The predictive framework also offers a transferable model for health and environmental applications in resource-limited settings.

Methods

Study setting and period

This study was conducted in 34 sub-Saharan African countries using the most recent Demographic and Health Survey (DHS) from 2012 To 2024, comprising the Household Record dataset (HR file). The countries that are included in this study are Angola, Benin, Burkina Faso, Burundi, Cameroon, Chad, Comoros, Congo Republic, Cote deivoirs, Ethiopia, Gabon, Gambia, Ghana, Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Senegal, Sierra Leone, South Africa, Tanzania, Togo, Uganda, Zambia, and Zimbabwe. This region

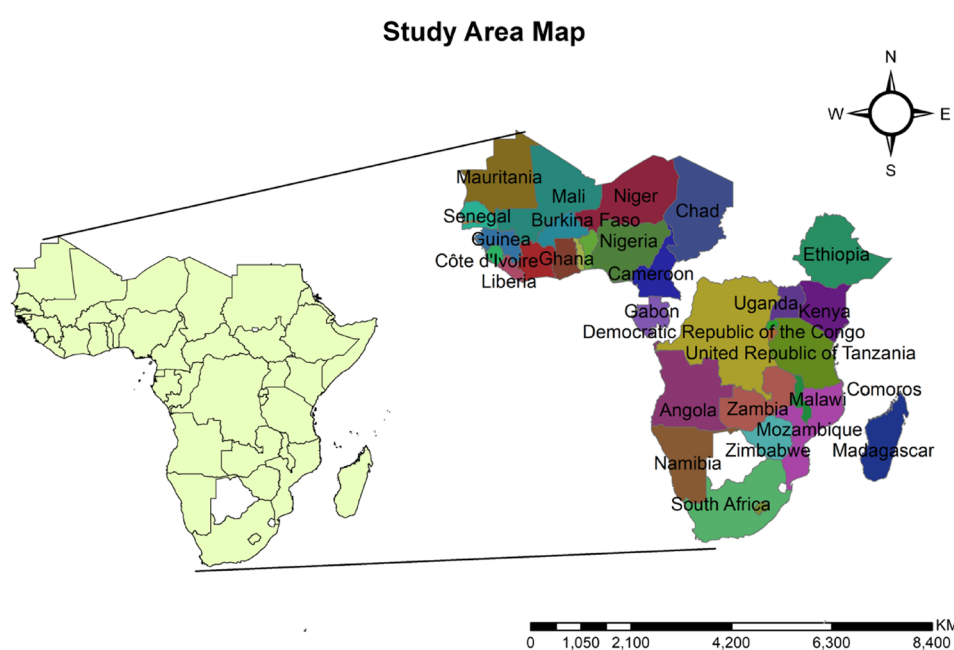
includes savanna, rainforests, and mountain ranges. Approximately forty-nine countries, including Nigeria, Ethiopia, Kenya, and South Africa, make up this region that stretches over 9.3 million square miles (24 million square kilometers) in area [31, 32]. Recently, it was estimated that the population exceeded 1.1 billion people, out of which a large percentage is below the age of 20 [33] (Fig. 1).

Data source and study population

For the analysis, we used the household recode (HR) of DHS data, combining data from 34 countries in sub-Saharan Africa. A standardized dataset was employed [34]. The purpose of Demographic and Health Surveys (DHS) is to collect a large, representative sample from the source population while accounting for relevant factors. To get comparable data on a global scale, these surveys use large sample sizes that are nationally representative and based on the population of each nation [34]. A total weighted sample size of 500,845 was considered for this study. The survey years vary across countries, ranging from 2012 To 2024. The largest sample size was from Nigeria (40,427), while the smallest was from Comoros (4,482) (Table 1). The data used in this study is publicly available and can be accessed at <https://dhsprogram.com/data/available-datasets.cfm>.

Sample size determination and sampling method

The DHS Program, in general, uses a standardized, stratified two-stage cluster sampling design to ensure



Source of shape file: Global Administrative Areas Database (GADM), own map output from ArcGIS V.10.7

Fig. 1 Map of sub-Saharan Africa, shape file for the study area map was taken from GADM and plotted using ArcGIS V. 10.7 software

Table 1 Study period and sample size for 34 sub-Saharan African countries included in this study

Country	Recent DHS year	Number of clusters	Weighted sample size
Angola	2015/16	627	16,109
Benin	2017/18	555	14,156
Burkina Faso	2021	600	13,251
Burundi	2016/17	554	15,977
Cameroon	2022	469	11,710
Chad	2014/15	626	17,233
Comoros	2012	252	4,482
Congo Republic	2013/14	540	36,342
Cote d'Ivoire	2021	539	14,766
Ethiopia	2016	645	16,650
Gabon	2019/21	390	11,781
Gambia	2019/20	281	6,549
Ghana	2022	618	17,933
Guinea	2018	401	7,912
Kenya	2022	1,692	37,911
Lesotho	2023/24	400	9,810
Liberia	2019/20	325	9,068
Madagascar	2021	657	20,510
Malawi	2015/16	850	26,361
Mali	2018	379	9,510
Mauritania	2019/21	1210	11,658
Mozambique	2022/23	619	14,250
Namibia	2013	554	9,849
Niger	2012	480	10,750
Nigeria	2018	1,400	40,427
Rwanda	2019/20	500	12,949
Senegal	2023	400	8,423
Sierra Leone	2019	578	13,399
South Africa	2016	750	11,083
Tanzania	2022	629	15,705
Togo	2013/14	330	9,549
Uganda	2016	697	19,588
Zambia	2018	545	12,831
Zimbabwe	2015	400	10,534
Total		20,492	500,845

nationally representative data. Within each administrative geographic region, the samples used for health and demographic surveys are usually separated into rural and urban regions. Enumeration Areas (EAs) were selected in the first sample round using a probability corresponding to their size within each stratum. In the second step of sampling, a predetermined number of homes from the targeted EAs were chosen using the systematic sampling approach. Equal probability systematic sampling was used after the household listing to choose a specific number of homes from the designated cluster [35]. This method ensures representation at national, urban-rural, and regional levels. In each selected household,

the mother or father was interviewed with an individual questionnaire.

Variables of the study

Outcome variable

The dependent (outcome) variable was the household sanitation facility access. Improved facility status was granted to households that accessed flush/pour flush to piped sewer systems, septic tanks or pit latrines, ventilated improved pit latrines, composting toilets, or pit latrines with slabs. The 'Unimproved' facility was used to categorize households using pit latrines without slabs or platforms, bucket latrines, and disposal of feces in fields, forests, bushes, open bodies of water, or other open spaces, or with solid waste [36]. The DHS data contains a variable entitled "Type of toilet facility," so we recorded those households' responses into two categories. Those households that used "Improved sanitation facility" as "0" and those households that used "Unimproved sanitation facility" were coded as "1".

Explanatory variables

The study included different household characteristics such as the age of the household head, sex of the household head, educational level, marital status, wealth index, number of household members, and number of children under 5. In addition, water source, location of water source, access to a water source, cooking fuel choice, access to electricity, household water treatment, shared toilets, hand-washing facilities, soap presence, place of residence, and household ownership of radio and television.

Machine learning model

Machine learning (ML) is the ability of a system to automatically acquire, integrate, and then develop knowledge from large-scale data, and then expand the acquired knowledge autonomously by discovering new information, without being specifically programmed to do so [37]. Machine learning is revolutionizing healthcare and public health, enabling data-driven insights through predictive analytics, pattern recognition, and automated complex processes [38, 39]. Machine learning, utilizing algorithms like neural networks, decision trees, and support vector machines to analyze vast datasets, has significantly impacted healthcare by enhancing diagnostics, treatment planning, operational efficiency, and assisting in clinical decision-making through pattern identification and outcome prediction [40].

Machine learning is crucial in public health for predictive analytics, surveillance, and outbreak prediction by analyzing population data to forecast disease spread, optimize resource allocation, and target interventions for at-risk groups [41, 42]. Machine learning enhances

healthcare and public health by delivering tailored, equitable care, improving outcomes, and reducing disparities [43].

Data management and analysis procedure

Data extraction was performed using STATA version 17, after which the data were converted into CSV format and imported into Python version 3.12 in Jupiter Notebook for further analysis. In this study, descriptive statistics were used to summarize sociodemographic characteristics, socioeconomic characteristics, exposure to mass media, and household and environmental characteristics through frequency and percentage analysis. To predict a household's sanitation facility, this study adopts a general

framework from a previous study, based on Yufeng Guo's 7 Steps of Machine learning [44]. These steps include data collection, preparation, model selection, training, evaluation, parameter adjustment, and prediction, applicable to both supervised and unsupervised machine learning. For this project, machine learning models and algorithms were implemented using the Scikit-learn Python module [45]. The data processing and analysis involved multiple stages, including data preprocessing, transformation, integration, dimensionality reduction, splitting, discretization, and feature engineering, ensuring a high-quality dataset for machine learning predictions (Fig. 2).

For data splitting and cross-validation, the full dataset was randomly stratified and split into a hold-out test set

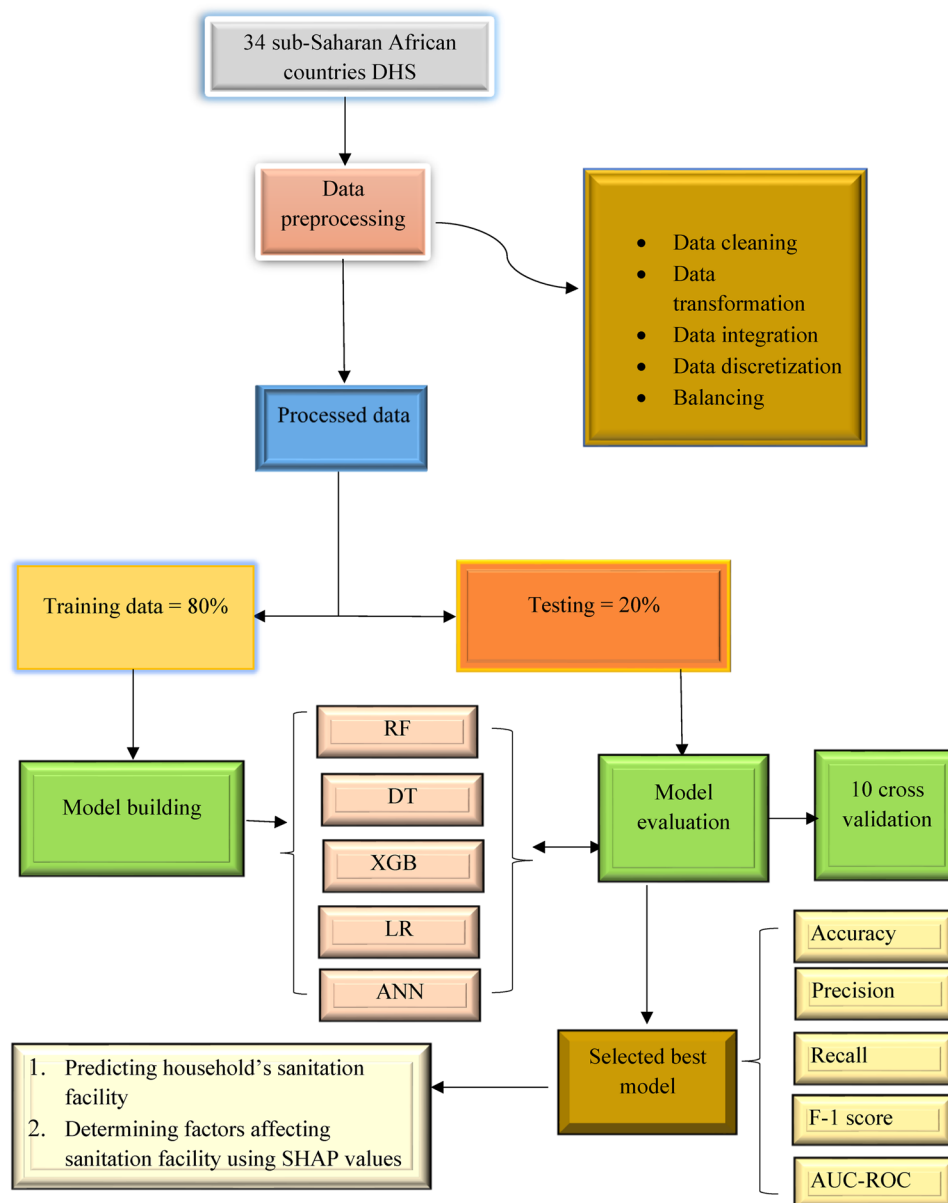


Fig. 2 Overall data preparation and analysis process for predicting household sanitation facilities in sub-Saharan Africa

(20%) and a training set (80%). All model development, hyperparameter tuning, and internal validation were performed using 10-fold cross-validation on the 80% training set. Within each fold, preprocessing and SMOTE balancing were applied only to the training partition to avoid data leakage, and models were evaluated on the fold's validation partition. For model selection, five supervised machine learning algorithms, such as Random Forest (RF), Decision Tree (DT), Extreme Gradient Boosting (XGB), Logistic Regression (LR), and Artificial Neural Network (ANN), were trained. Both Decision Tree (DT) and Random Forest (RF) models were included to balance interpretability and predictive performance. Model selection was based on the mean cross-validated performance metrics (accuracy, precision, recall, F1, AUC-ROC). The model with the best predictive performance was selected for final analysis. After selecting the optimal algorithm and hyperparameters, the final model was retrained on the entire 80% training set, and its performance was reported on the independent 20% hold-out test set.

Feature importance was assessed using Shapley Additive Explanations (SHAP), which estimates each variable's contribution to the model's prediction. This approach quantifies the relative influence of each feature on sanitation facility prediction rather than relying on traditional statistical significance testing. Furthermore, association rule mining was applied using the Apriori algorithm to detect frequently occurring features that significantly impact a household's sanitation facility.

Both Decision Tree (DT) and Random Forest (RF) models were included to balance interpretability and predictive performance. The DT algorithm was used as a transparent baseline model that allows visualization of decision paths and identification of key predictor variables affecting sanitation facility utilization. In contrast, RF, as an ensemble of multiple decision trees built through bootstrapped sampling and random feature selection, was included to assess whether ensemble learning could improve generalization and predictive accuracy while mitigating over-fitting commonly associated with single-tree models.

Data preprocessing

Data processing is a machine learning technique that transforms raw data into an understandable format [46]. It is the first step in machine learning, ensuring a clean, consistent, and structured dataset. It involves handling missing values, removing duplicates, addressing outliers, standardizing formats, and encoding categorical variables. In this study, we employed four key data processing techniques: data cleaning, data integration, data transformation, and discretization. Data processing and analysis are performed using Python (version 3.9) software and

some basic packages like Pandas, Scikit-Learn, Imblearn, Theano, Numpy, and Seaborn or Matplotlib, which are utilized for preparing data, discretizing data, transforming data, choosing, visualizing, exploring, training, and evaluating models.

Data cleaning

Data cleansing is the initial step after data retrieval, focusing on correcting outcome variable categories, addressing missing values, and identifying and removing outliers. In this study, the k-nearest neighbors (KNN) imputation technique was applied to estimate missing values for the independent variables. Previous research has shown that KNN imputation is a more reliable and sensitive method for handling missing data [47].

Data transformation

Due to the non-aggregate nature of the DHS dataset, it is necessary to reorganize or restructure the raw data and convert the data to create the same data types [48]. Data transformation involves converting data into a suitable format for analysis, which may include modifying data types, scaling, normalizing, or renaming variables [49]. Most variables in the dataset were categorical. These were transformed into numerical form using label encoding for ordinal variables (e.g., education level, wealth index) and one-hot encoding for nominal variables (e.g., sanitation type, residence, water source, and cooking fuel). Continuous numerical predictor variables, including household size, number of under-5 children, and age of household head, were scaled using StandardScaler from Scikit-Learn, which standardizes each variable to have a mean of 0 and a standard deviation of 1. This makes the data more appropriate for analysis, facilitating efficient model training, stability, comparability, and improved performance.

Data discretization

Data discretization is the process of converting continuous data into discrete categories or bins [50]. This approach is commonly used when certain models perform better with categorical inputs. Discretization can simplify the dataset, mitigate the influence of outliers, and enhance the interpretability of the model results. In the present study, continuous variables related to factors influencing sanitation facility access in sub-Saharan Africa were discretized to enable more meaningful analysis of risk levels or behaviors associated with sanitation practices [51, 52]. Specifically, household size was categorized as <4 or ≥ 4 members, number of children under five as 0, 1–2, or ≥ 3 , age of the household head as <35 , 35–60, or >60 years, and time to access a water source as <30 min or ≥ 30 min.

Data integration

Data integration involves combining data from different sources to create a unified dataset that can be used for analysis. This is particularly relevant in machine learning tasks where multiple datasets might be collected from various sources to build a more comprehensive model. Data integration helps to merge diverse features into one consistent dataset, ensuring that all relevant information is included for better predictions and insights [53, 54]. In this study, data integration was carried out by combining various datasets related to sanitation practices, demographic information, and environmental factors. This enabled the development of a comprehensive model for predicting sanitation facilities in Sub-Saharan Africa.

Class balancing

Class balancing is a technique used to address imbalances in the distribution of classes within a dataset, which can significantly affect the performance of machine learning models. When presented with an uneven data set, machine learning algorithms are prone to bias toward the majority class, leading to poor prediction accuracy for the minority class [55]. Several approaches can be used for class balancing, including under-sampling the majority class, oversampling the minority class, or using algorithms specifically designed to handle imbalanced datasets (e.g., class-weighted algorithms). In this work, we employed the synthetic minority oversampling technique (SMOTE) to improve model performance and ensure accurate predictions of sanitation facilities [56–58].

Model training

Model training is the process of teaching a machine learning algorithm to recognize patterns in data by adjusting its internal parameters to minimize prediction errors. This involves feeding the model with labeled training data, allowing it to learn relationships between input features and target variables [59]. In this study, the selected model was trained using a preprocessed dataset containing demographic, environmental, housing, and socioeconomic factors influencing sanitation facilities in sub-Saharan Africa. The dataset was split into 80% for training and 20% for testing to ensure unbiased learning and evaluation. A tenfold cross-validation method on the training data was used for training the models. The model is trained on nine of these folds and tested on the remaining fold, with this process repeating ten times so that each fold serves as a test set once. The final performance is then averaged across all iterations, providing a more reliable estimate of the model's generalizability. In this study, tenfold cross-validation was applied to evaluate the predictive performance of the selected model in

identifying sanitation facilities in sub-Saharan Africa [60].

Model selection

Model selection is a critical step in machine learning, involving the choice of the most appropriate algorithm to achieve optimal predictive performance. In this study, we evaluated several machine learning models to identify the best-performing approach for predicting household sanitation facility access in sub-Saharan Africa. The models considered included Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Artificial Neural Network (ANN), and Logistic Regression (LR).

Model performance was assessed using the test dataset. Since the primary outcome, household sanitation facility access, is categorical, classification performance was evaluated using the area under the receiver operating characteristic (ROC) curve, accuracy, sensitivity, and specificity. ROC-AUC was selected as the primary metric for model comparison because it provides a comprehensive measure of predictive accuracy regardless of the classification threshold [61]. An ROC value of 1 indicates perfect classification [62, 63]. Although mean absolute percentage error (MAPE) is commonly used for continuous outcomes, in this study, no continuous outcome variables were directly analyzed using MAPE. The evaluation focused entirely on classification metrics appropriate for the binary/categorical nature of the outcome. This approach ensures that model selection accurately reflects predictive performance in identifying households with or without access to improved sanitation facilities.

Performance metrics

Model evaluation metrics are essential for assessing the performance of machine learning models. These metrics vary based on the type of machine learning problem: classification, regression, or clustering [64]. The performance of the trained models was assessed using classification metrics, including accuracy, ROC curve, precision (P), recall (R), and F-1 score measure, and the best model was chosen.

T_Ps (True Positives) represent the number of correctly predicted positive instances.

T_Ns (True Negatives) represent the number of correctly predicted negative instances.

F_Ps (False Positives) represent the number of incorrectly predicted positive instances.

F_Ns (False Negatives) represent the number of incorrectly predicted negative instances.

Sensitivity (True Positive Rate or Recall): It is the capacity of a test to accurately identify patient cases [65]. Measures how many actual positives were correctly predicted.

This suggests that a different percentage of actual positive instances might be forecasted as negative (false negative) (Eq. 1).

$$\text{Sensitivity or Recall} = \frac{TP}{TP+FN}$$

Accuracy: is the ability to differentiate the case and control correctly [65]. It is calculated as the proportion of true positives and true negatives in all evaluated cases (Eq. 2).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision (Positive Predictive Value): Measures how many of the predicted positives are positive (Eq. 3).

$$\text{Precision} = \frac{TP}{TP+FP}$$

F1 score: Harmonic mean of precision and recall; useful when classes are imbalanced (Eq. 4).

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

ROC-AUC (Receiver operating Characteristic-Area under the Curve): Measures the model's ability to distinguish between classes at various threshold levels [66]. The model with the highest ROC curve and accuracy estimates was selected.

Hyper-parameter tuning

To address potential overfitting and optimize model performance, we implemented the 10-fold cross-validation approach [67]. This procedure divides the data into ten subsets, iteratively cycling through each subset while using one subset for validation and the remaining nine for training. For each model, the hyperparameters were adjusted to maximize the receiver operating characteristic (ROC) curve and F1-score, both of which are common evaluation metrics for binary classification tasks. This process was repeated five times with different training and testing splits to ensure robustness and reduce the risk of overfitting.

In addition, the Random Forest (RF) model, identified as the best-performing classifier, underwent further hyperparameter optimization using GridSearchCV with 5-fold cross-validation and the F1-score as the selection criterion. The search space included: `n_estimators` [100, 200, 500], `max_depth` [None, 10, 20, 30], `min_samples_split` [2, 5, 10], `min_samples_leaf` [1, 2, 4]. The grid search was performed on the resampled (SMOTE-balanced) training dataset. The optimal parameter combination that achieved the highest mean cross-validated F1-score was selected and applied in the final Random Forest model. This process enhanced the model's predictive performance, reduced overfitting, and ensured reproducibility across analyses.

Model explanation and feature selection

In this study, Shapley Additive Explanations (SHAP) values were used to explain the contribution of individual features to the model's predictions. While the RF model highlights feature significance in predicting the density of a household's sanitation facility, it falls short of detailing the influence of these features. SHAP steps in to fill this interpretive gap by quantifying the impact of independent variables on the dependent variable. SHAP values, based on cooperative game theory, assign a value to each feature, representing its impact either positive or negative on the prediction [68, 69]. It is often favored over other approaches it offers interpretability that is both consistent and independent of any specific model and employs Shapley values to quantify the impact of each feature in a prediction model [70]. This approach not only facilitates a global understanding of the model's behavior but also enables detailed local interpretations, which are essential for understanding complex phenomena. It calculates the average marginal contribution of each feature to the prediction outcome. Features with high SHAP values indicate those that are most influential in predicting the target variable, which is the sanitation facility. The Shapley value for a given feature i is defined as the weighted sum of its marginal contributions across all possible combinations of features, formulated as (Eq. 5):

$$\phi_i = \sum_{S \subseteq N, i \in S} \frac{|S|(n-|S|-1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

Here, $|S|$ signifies the number of features in a subset S , n is the total count of features, $v(S)$ is the model prediction with the current set of features, and $v(S \cup \{i\})$ is the prediction when feature i is added to the subset S [61]. By applying this method, we identified the specific contributions of each independent variable to the prediction of household sanitation facility preference, providing critical insights that are instrumental for sanitation planning and policy-making.

Results

Sociodemographic characteristics of the study participants

Generally, 500,845 household samples were collected from countries in sub-Saharan Africa. Of the participants, almost 71% (71.24%) of household heads were men. More than half of the household heads (52.38%) were between 35 and 60 years of age. More than 34% (34.46%) lacked any formal schooling, whereas more than 72% (72.62%) of the participants were married. 49.06% of households have more than 4 family members, while 45.85% of households have no children under 5 in their dwelling. Among all participants, more than 22% (22.11%) of the households had the lowest wealth index (poorest). Regarding their place of residence, around 63% (62.88%) reside in rural parts of the nation (Table 2).

Table 2 Sociodemographic-related characteristics of households in sub-Saharan Africa using DHS

Variables	Category	Frequency (%)	Sanitation facility (%)	
			Improved (%)	Unimproved (%)
Age of household head	< 35	145,168 (28.98)	72,583 (14.49)	72,585 (14.49)
	35–60	262,343 (52.38)	133,501 (26.66)	128,842 (25.72)
	> 60	93,334 (18.64)	45,070 (8.99)	48,264 (9.64)
Sex of household head	Male	356,778 (71.24)	177,353 (35.41)	179,425 (35.82)
	Female	144,067 (28.76)	73,801 (14.74)	70,266 (14.03)
The highest educational level attained	No education	172,602 (34.46)	61,263 (12.23)	111,339 (22.23)
	Primary	156,419 (31.23)	74,101 (14.79)	82,318 (16.44)
	Secondary	130,059 (25.97)	80,937 (16.16)	49,122 (9.81)
	Higher	41,765 (8.34)	34,853 (6.96)	6,912 (1.38)
Marital statuses	Never married	41,570 (8.30)	26,296 (5.25)	15,274 (3.05)
	Married	363,689 (72.62)	179,068 (35.75)	184,621 (36.86)
	Widowed	60,515 (12.08)	28,894 (5.77)	31,621 (6.31)
	Divorced	16,896 (7.00)	18,175 (3.63)	108,181 (21.60)
Wealth index combined	Poorest	110,730 (22.11)	55,582 (11.09)	55,148 (11.01)
	Poorer	98,356 (19.64)	49,010 (9.79)	49,346 (9.85)
	Middle	98,555 (19.68)	49,507 (9.88)	49,048 (9.79)
	Richer	97,099 (19.39)	48,614 (9.71)	48,485 (9.68)
	Richest	96,105 (19.19)	48,441 (9.67)	47,664 (9.52)
Number of households Members	<=4	255,126 (50.94)	134,901 (26.93)	120,225 (24.00)
	>4	245,719 (49.06)	116,253 (23.21)	129,466 (25.85)
Number of children under 5	No child	229,642 (45.85)	125,843 (25.13)	103,799 (20.72)
	1–2	232,417 (46.40)	110,464 (22.06)	121,953 (24.35)
	>=3	38,786 (7.74)	14,847 (2.96)	23,939 (4.78)
Place of Residence	Urban	185,924 (37.12)	93,613 (18.69)	92,311 (18.43)
	Rural	314,921 (62.88)	157,541 (31.46)	157,380 (31.42)

Housing and environmental characteristics of the study participants

Over 30% (30.41%) of all households use unimproved drinking water sources, and 90.76% of households get their water outside of their premises (elsewhere). More than 80% (80.17%) of households do not use any kind of household water treatment, and more than 80% of the participants (18.25%) of dwellings have limited access to water sources. 59.14% of the houses have access to electricity, while a small percentage (15.71%) of households use clean fuel as a source of energy. More than half of the participants (52.48%) have no radio, and 66.32% of families have no television in their house. 53.88% of the households use shared toilet facilities. Of all participants, 30.93% of households have no hand washing facility in their dwellings, while 75.68% of households have soap or detergent near hand washing facilities (Table 3).

The distribution of sanitation facilities across sub-Saharan African countries

The findings of this study revealed that 49.85% (95% CI (49.71%, 49.99%)) of the households use unimproved sanitation facilities, while 50.14% (95% CI (50.01%, 50.28%)) of households use improved sanitation facilities (Fig. 3). The highest proportion of unimproved sanitation users was observed in Congo (51.22%), South Africa (51.14%), and Madagascar (50.66%). In contrast, the smallest proportion was observed in Gambia (49.06%) (Fig. 4).

Balancing the imbalanced dataset

Before balancing, the dataset exhibited a slight class imbalance, with 200,893 households using improved sanitation facilities (Class “0”) and 199,783 households using unimproved sanitation facilities (Class “1”). After applying SMOTE, the dataset was balanced, resulting in 200,893 instances for each class. This adjustment improved the model’s ability to learn from both groups and reduced bias in predicting household sanitation facility access across sub-Saharan Africa (Fig. 5).

Effect of sampling techniques on model performance

To assess the impact of imbalance-handling methods on model performance and potential overfitting, a comparative analysis was conducted between the Synthetic Minority Oversampling Technique (SMOTE) and undersampling. Overall, SMOTE yielded slightly higher F1-scores and AUC values, particularly for the Random Forest and Logistic Regression models, indicating a modest improvement in classification balance and discriminatory power. The XGBoost and Artificial Neural Network (ANN) models demonstrated comparable performance across both techniques, suggesting stability regardless of sampling strategy. In contrast, the Decision Tree model exhibited a lower AUC when using undersampling (0.7984) compared to SMOTE (0.8345), although its F1-scores remained relatively consistent (0.7864).

Table 3 Housing and environmental-related characteristics of households in sub-Saharan Africa using DHS

Variables	Category	Frequency (%)	Sanitation facility (%)	
			Improved (%)	Unimproved (%)
Drinking water source	Unimproved	152,315 (30.41)	48,246 (9.63)	104,069 (20.78)
	Improved	348,530 (69.59)	202,908 (40.51)	145,622 (29.07)
Location of water source	In the premises	8,017 (1.60)	5,779 (1.15)	2,238 (0.45)
	In own Yard	38,243 (7.64)	22,325 (4.46)	15,918 (3.17)
	Elsewhere	454,585 (90.76)	223,050 (44.53)	231,535 (46.23)
Household water treatment	No	401,544 (80.17)	192,202 (38.38)	209,342 (41.79)
	Yes	99,301 (19.83)	58,952 (11.77)	40,349 (8.06)
Access to a water source	Basic access (<= 30 min)	409,455 (81.75)	219,818 (43.89)	189,637 (37.86)
	Limited access (> 30 min)	91,390 (18.25)	31,336 (6.26)	60,054 (11.99)
Access to electricity	No	296,204 (59.14)	100,079 (19.98)	196,125 (39.16)
	Yes	204,641 (40.86)	151,075 (30.16)	53,566 (10.69)
Has radio	No	257,825 (52.48)	108,535 (21.67)	149,290 (29.81)
	Yes	243,020 (48.52)	142,619 (28.48)	100,401 (20.04)
Has television	No	332,151 (66.32)	123,850 (24.73)	208,301 (41.59)
	Yes	168,694 (33.68)	127,304 (25.42)	41,390 (8.26)
Cooking fuel choice	Clean fuel	78,693 (15.71)	64,001 (12.78)	14,692 (2.93)
	Solid fuel	422,152 (84.29)	187,153 (37.37)	234,999 (46.92)
Shared toilet	No	231,003 (46.12)	198,721 (39.68)	317,288 (63.35)
	Yes	269,842 (53.88)	16,653 (3.32)	221,412 (44.21)
Hand washing facility	Yes	345,951 (69.07)	189,137 (37.76)	156,814 (31.31)
	No	154,894 (30.93)	62,017 (12.38)	92,877 (18.54)
Presence of soap/detergent near the hand washing facility	No	379,050 (75.68)	167,411 (33.42)	211,639 (42.26)
	Yes	121,795 (24.32)	83,743 (16.72)	38,052 (7.59)

vs. 0.8242) (Fig. 6). These findings suggest that while SMOTE provides a minor performance advantage for certain models, the differences are not substantial. Consequently, undersampling remains a viable alternative, particularly for reducing the risk of overfitting by avoiding synthetic data generation.

Machine learning classifier results

Model performance metrics are reported both on the independent test set and using 10-fold cross-validation to provide robust estimates of generalization. Cross-validated accuracy was slightly higher than raw test-set accuracy for all models, indicating stable performance across different folds. The cross-validation results were used as the primary reference for comparing model robustness and predictive capability. Random Forest performed best overall, with the highest Accuracy (0.8061) and F1-score (0.8377), indicating strong predictive power and balance between precision and recall. Decision Tree had the highest Precision (0.9089), meaning it minimized false positives, and the Recall value of (0.7539) suggests it missed some actual cases. The remaining models (Logistic Regression, XGBoost, and ANN) showed similar performance, with high precision (0.8929, 0.8922, 0.8937) and F1 scores of (0.7851, 0.8145, 0.8079), respectively, demonstrating reliability in classification. Overall, all models performed well, with Random Forest being the most robust for predicting sanitation facilities in sub-Saharan Africa (Table 4).

ROC analysis showed all models effectively predicted sanitation facilities in sub-Saharan Africa, with Decision Tree, XGBoost, and ANN performing best (AUC=0.83) while Logistic Regression (AUC=0.81) and Random Forest (AUC=0.86) also performed well, all surpassing random chance (AUC=0.5) (Fig. 7). Overall, the Random Forest model demonstrated the highest accuracy (0.86) among the evaluated machine learning models, along with a strong F1-score (0.8377), indicating a well-balanced performance between precision and recall. This suggests that the model effectively identifies both households that use unimproved and improved sanitation facilities, minimizing misclassification errors. Given its superior accuracy and robust predictive capabilities across multiple evaluation metrics, Random Forest emerges as the best-performing model for predicting sanitation facilities in sub-Saharan Africa. Random Forest is an ensemble algorithm that combines multiple decision trees and makes final predictions based on majority voting [71].

Decision tree vs. Random forest comparison

Although Decision Tree (DT) and Random Forest (RF) share similar algorithmic foundations, the inclusion of both models is justified: the DT represents a single-tree

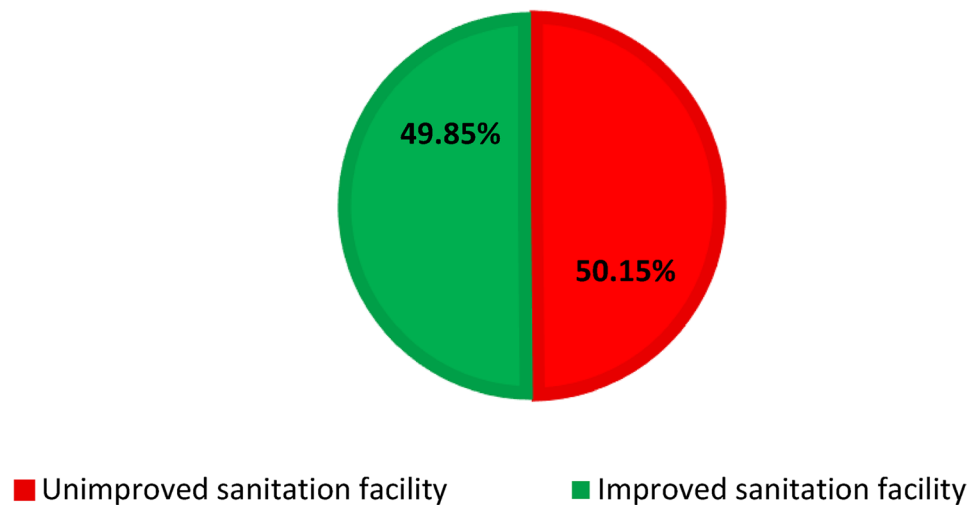


Fig. 3 The proportion of the type of sanitation facility utilized by households in Sub-Saharan Africa

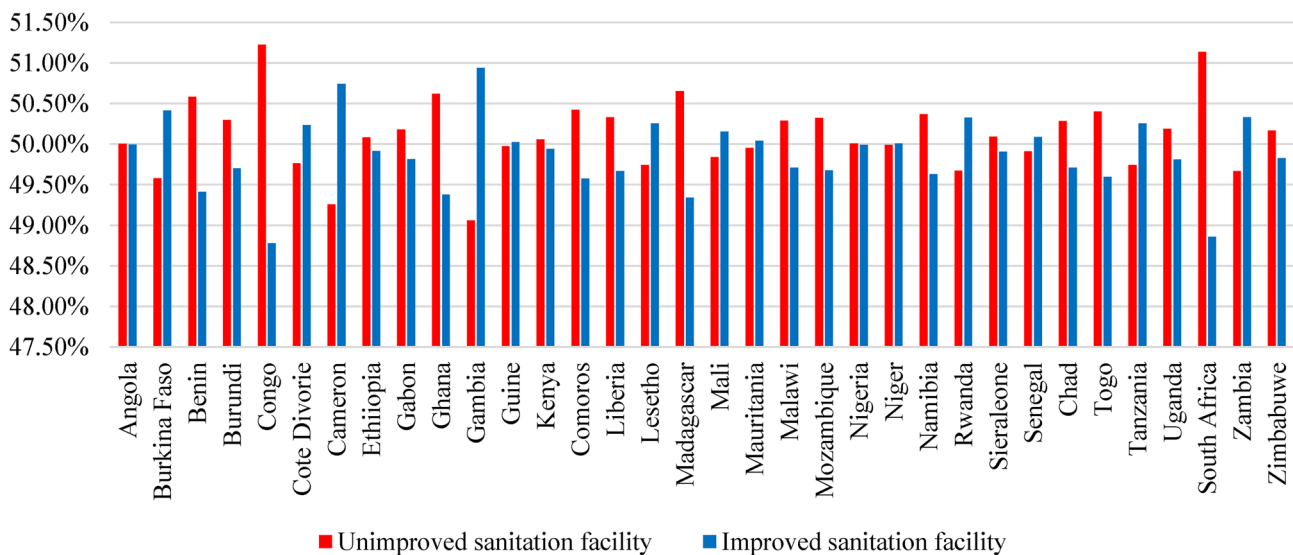


Fig. 4 The proportion of types of sanitation facilities utilized across countries in Sub-Saharan Africa

baseline, while RF is an ensemble method that aggregates multiple trees to improve predictive performance and reduce variance. To assess whether the ensemble approach provides a statistically significant improvement, McNemar's test was conducted to compare the error rates of DT and RF on the test set. The test yielded a statistic of 3107.0 with a p-value < 0.001, indicating a significant difference in error rates between the two models (Table 5). These results confirm that Random Forest significantly outperforms the single Decision Tree model, supporting its selection as the best-performing classifier.

Determinant factors for a household's sanitation facility utilization

Based on the feature-importance analysis, the y-axis in Fig. 8 ranks the variables in order of their

relative importance, while the x-axis shows the normalized importance values ranging from 0.0 to 1.0. The variable with the highest score (set to 1.0) serves as the reference point, indicating the greatest contribution to the model's prediction. All other variables are expressed as proportions of this top feature's influence, meaning that values smaller than 1.0 still represent meaningful though relatively lower levels of importance.

Accordingly, shared toilet stands out as the most influential predictor (0.233), followed by education (0.204) and wealth index (0.094). Residence of the household (0.067) also shows a notable effect. Variables such as access to electricity (0.038), age of the household head (0.037), drinking-water source (0.036), household water treatment (0.033), and cooking-fuel choice (0.031) demonstrate moderate importance, contributing

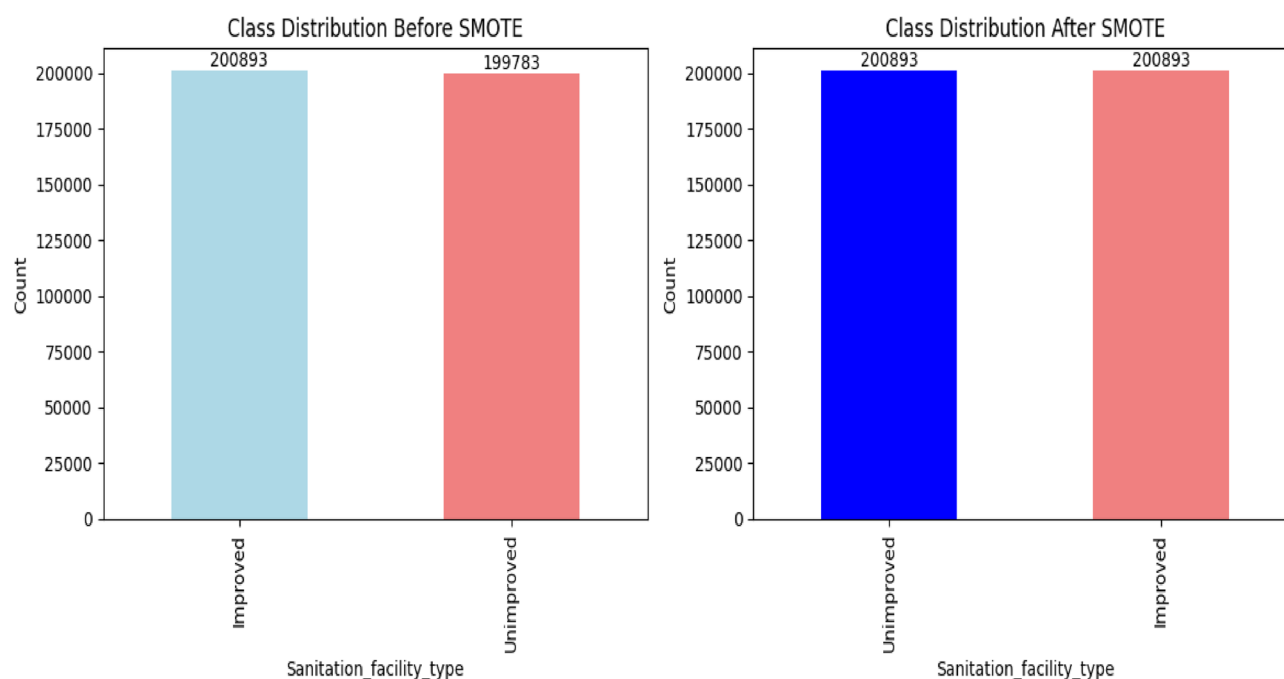


Fig. 5 Class distribution of sanitation facilities in sub-Saharan Africa before and after smote balancing

meaningfully to sanitation-facility utilization. In addition, access to radio (0.029) and access to television (0.027) indicate that media exposure may play a supporting role. Factors including sex of the household head (0.027), hand-washing facility (0.025), water access (0.021), soap presence (0.021), location of the water source (0.020), marital status (0.020), and number of household members (0.020) exhibit smaller but still measurable effects. Finally, the number of under-five children (0.017) appears to exert the least influence on the model's predictions.

SHAP value

The SHAP (Shapley Additive Explanations) summary plot presents the relative contribution of each feature to the model's predictions of sanitation facility types among households in sub-Saharan Africa (Fig. 9). Each dot represents a household, with colors showing the magnitude of the feature value (red = high, blue = low). Features are ordered by their average contribution to the model's output, with those at the top having the greatest influence.

Among all features, shared toilet, education level, and wealth index had the largest contributions to the model's predictions. Households using shared toilets were more frequently associated with predictions of unimproved sanitation, whereas higher education levels and greater wealth were associated with predictions of improved sanitation. Residence (urban/rural) also played an influential role; rural households were more often linked with model predictions of unimproved sanitation facilities. Access to mass media (radio and television) contributed to predictions of improved sanitation, suggesting that media

exposure may be an important contextual indicator of awareness and socioeconomic advantage.

Other features, such as hand-washing facility, household water treatment, drinking water source, household water access, and location of water source, also contributed to the model, indicating that better access to water and hygiene resources was generally associated with predictions of improved sanitation. Socio-demographic features, including age and sex of the household head, marital status, household size, and number of children under five years old, were also informative to the model's predictions. For instance, households with older heads and smaller family sizes tended to be linked with model predictions favoring improved sanitation. Finally, cooking fuel, electricity, and the presence of soap at the washing area appeared as moderate but relevant predictors, reflecting broader household living standards.

It is important to emphasize that SHAP values do not represent causal effects or regression coefficients. Instead, they describe how much each feature contributed to the model's prediction for a given observation relative to a baseline. Therefore, these findings reveal the features most influential in shaping the model's predictive behavior, not cause-and-effect relationships between household characteristics and sanitation outcomes.

Assessment of predictor collinearity

We evaluated potential collinearity among key predictors, particularly socioeconomic-related variables such as Wealth index, Electricity access, and Radio ownership, given their potential influence on SHAP value

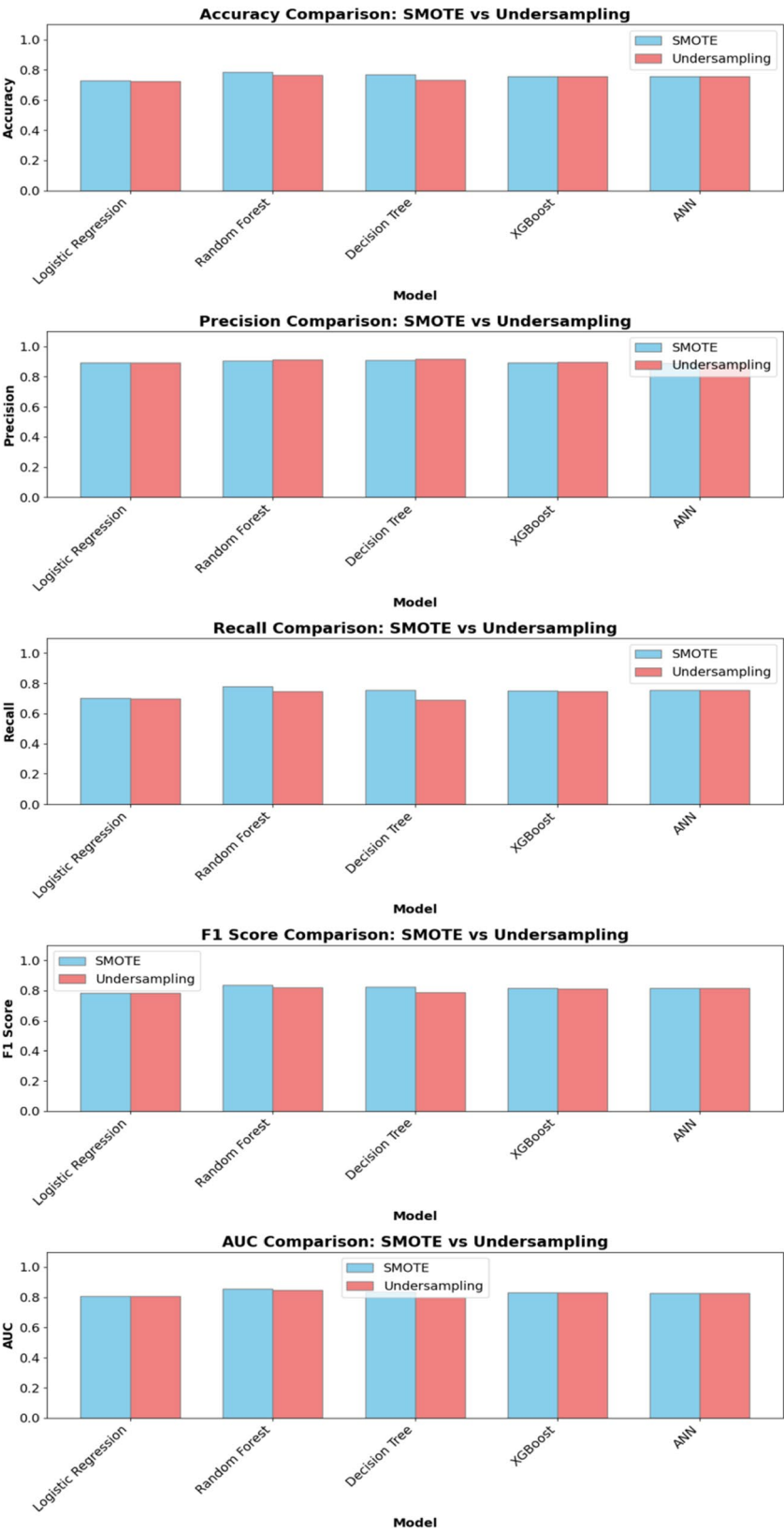


Fig. 6 Comparison of model performance using SMOTE and under-sampling techniques for balancing class distribution

Table 4 Performance metrics of machine learning models for predicting sanitation facility access in sub-Saharan Africa

Model	Test Set Accuracy (%)	10-fold CV Accuracy (95% CI)	Precision	Recall	F1 Score	AUC
Logistic Regression	0.7266	0.7691 (0.7478–0.7905)	0.8929	0.7005	0.7851	0.8061
Random Forest	0.7843	0.8061 (0.7644–0.8477)	0.9034	0.7809	0.8377	0.8560
Decision Tree	0.7707	0.7971 (0.7489–0.8453)	0.9089	0.7539	0.8242	0.8345
XGBoost	0.7567	0.7734 (0.7501–0.7967)	0.8922	0.7493	0.8145	0.8305
ANN	0.7502	0.7726 (0.7495–0.7958)	0.8937	0.7372	0.8079	0.8270

Test set accuracy (%) refers to raw accuracy on the independent test set, showing how the model performed on held-out data. 10-fold CV Accuracy (95% CI) is a cross-validated accuracy that is a more robust estimate with reduced sampling bias

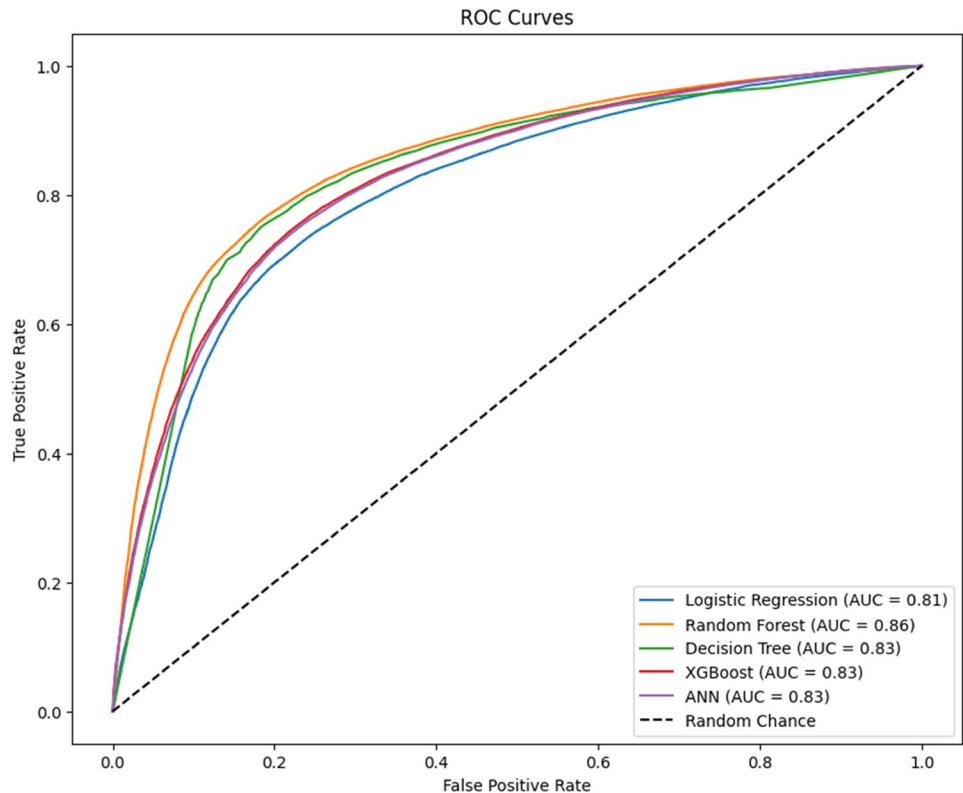


Fig. 7 Receiver operating characteristic (ROC) curves for machine learning models

Table 5 Summary table of the statistical significance testing conducted to compare the performance of the decision tree and random forest models

Models Compared	Decision Tree, Random Forest
Statistical Test Used	McNemar's Test (for comparing paired binary classification models' error rates)
Data Used for Testing	Predictions on the held-out Test Set
McNemar's Test Statistic	3107.0000
McNemar's P-value	0.0000
Significance Level (α)	0.05
Statistical Significance	Yes (P-value < α)

interpretation. A correlation analysis revealed a moderate positive correlation between Wealth index and Electricity ($r=0.55$), and weaker positive correlations between Wealth index and Radio ($r=0.32$) and between Electricity and Radio ($r=0.23$). None of the pairwise

correlations exceeded commonly accepted thresholds for strong collinearity ($r>0.7$ – 0.8), suggesting that multicollinearity is unlikely to substantially distort model interpretation. These findings support the reliability of the feature importance results obtained from SHAP values (Fig. 10).

Discussion

This study aimed to predict household sanitation facility access in sub-Saharan Africa using machine learning models from recent standard DHS data of each country. Currently, there appears to be no published study that specifically employs machine learning techniques to predict sanitation facility access in sub-Saharan Africa. The study demonstrates the efficacy of various machine learning models, with Random Forest emerging as the best-performing model, achieving the highest

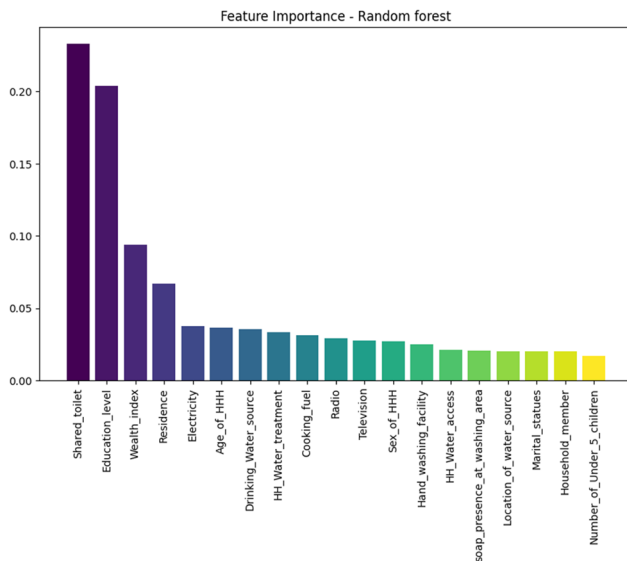


Fig. 8 Feature importance of variables associated with sanitation facility access using the Random forest model

accuracy (0.8061) and F1 score (0.8377). This suggests that Random Forest is particularly well-suited for handling complex interactions among the socioeconomic and demographic, housing, and environmental variables influencing household sanitation utilization. Therefore, this study can enhance policy formulation, optimize

resource allocation, and support SSA countries' efforts to achieve Sustainable Development Goal 6 Target 2 to prevent human interactions with fecal pathogens to reduce incidences of diseases by 2030 [27].

The proportion of unimproved sanitation facility access among households in SSA was 49.06% (95% CI (49.71%, 49.99%)). This finding was lower than studies conducted in rural areas in Ethiopia (56%) [72] and Ethiopia (67%) [73]. The lower proportion can be attributed to the development of effective sanitation initiatives, such as the Community-Led Total Sanitation (CLTS) programs. The greater emphasis of SDGs' other targets towards increasing international collaboration and capacity-strengthening support to less economically developed countries in water and sanitation-associated actions and projects can also contribute to a lower proportion [74]. Urbanization, stronger government and international support, education level, income, improvement in the communities' attitudes, and targeted public health campaigns may have further reduced unimproved sanitation facility access in SSA recently. Our findings are higher than those reported for Ethiopia (21.7%) [75], East African cities (40.3%) [2], and Benin (32%) [76]. The explanation for this could be due to the difference in sample size because our study includes 34 SSA, which might also increase the proportion.

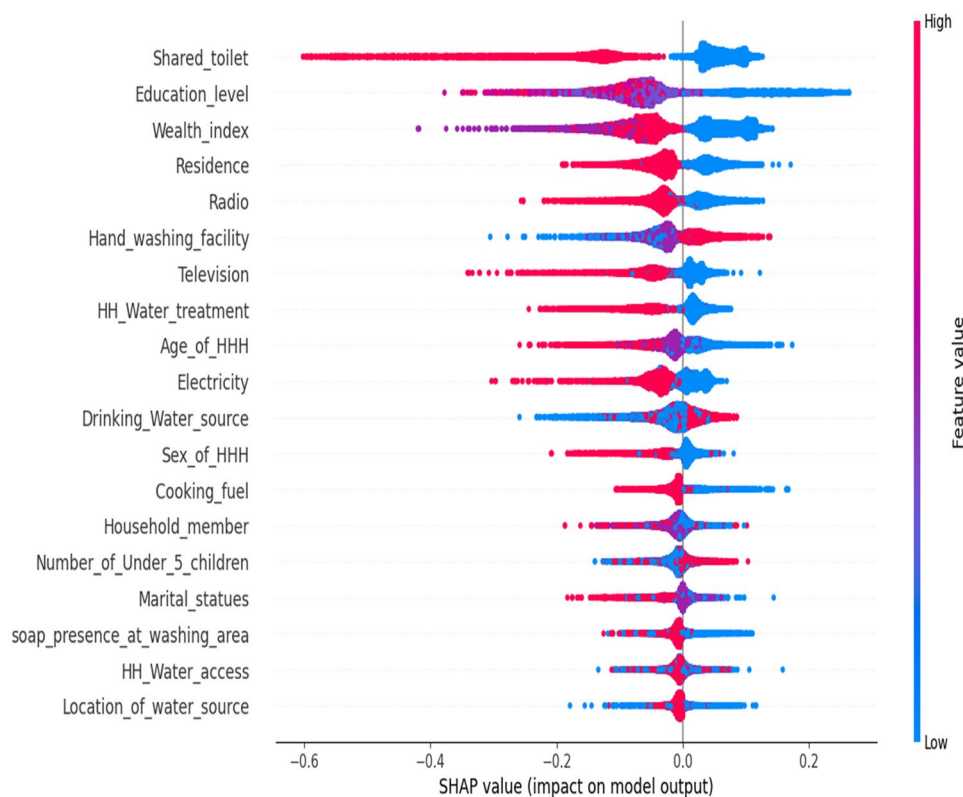


Fig. 9 SHAP summary plot: key predictors of household sanitation facility access in Sub-Saharan Africa



Fig. 10 Correlation matrix of key predictors of household sanitation facility access in sub-Saharan Africa

Our study from feature importance analysis revealed that the shared toilet was the strongest predictor of sanitation facility utilization. These findings are consistent with existing literature conducted in Ethiopia [77], which highlights the use of shared facilities as a greater contributor to unimproved sanitation facility access. Even if shared toilets are designed to be an improved type of sanitation facility, they are considered unimproved because they are shared among multiple users [78]. Another study in Kenya [79], Western Kenya [80], and Tanzania [81] stated that although most shared facilities had complete superstructures, they were not hygienically clean, as evidenced in their hygiene and slab aspects, which is the most challenging aspect of these sanitation facilities. Furthermore, they can lead to the spread of infections due to poor hygiene, limited accessibility, and unsafe conditions [82–85]. These are frequently not well kept, creating unclean conditions that discourage frequent use [86]. Research from throughout the world has linked the usage of shared restrooms to negative health consequences such as fecal-oral illnesses, diarrhea, enteric fevers, and helminth infections [87, 88]. Due to the above-indicated reasons, it aids the transmission of microorganisms that cause diarrheal diseases.

The feature importance analysis revealed that the education level was the stronger predictor of sanitation

facility utilization. This finding was supported by the studies conducted in Indonesia [89], Ghana [21], Ethiopia [90], and East Africa [2], which stated that more educated households used modern or improved sanitation facilities while less educated ones used unimproved pit latrines and practiced open defecation the most. A likely reason is that educated households are more aware of the risks of unimproved sanitation and poor hygiene associated with pit latrines, which leads to a decreased proportion of unimproved sanitation facility utilizers as well as the transmission of diseases such as feco-oral diseases. Educational attainment is thus key to understanding inequity in access to sanitation facilities.

In addition, wealth index was another major predictor of sanitation facility access following the education level. These findings are consistent with existing literature, which highlights poverty as a major determinant of unimproved sanitation facility access [91–95]. Poorer households often cannot afford to construct or maintain improved sanitation facilities such as pit latrines or flush toilets due to the cost of materials, labor, and even land space for toilets [96, 97]. Poor households that are lucky to have toilet facilities usually share with other households, whilst wealthy households have a better economy to provide improved toilet facilities for their families [93].

Residence also influences a household's sanitation facilities in SSA. Studies in Ethiopia, Nigeria's Osun State, and SSA stated that rural households were more likely to use unimproved sanitation facilities, mostly practicing open defecation, than urban ones [98–100]. Residence influences household sanitation facility access due to factors like inadequate sanitation infrastructure, economic constraints, cultural practices, and lack of awareness, especially in rural areas [99].

According to the results of our study, electricity access has a significant impact on a household's sanitation facility utilization. The finding is supported by the findings of a study conducted in Yemen [101] and Nigeria [7], which suggests households that have electricity access have a high chance of using improved sanitation facilities as compared to others. The possible explanation for this could be that some sanitation facilities, such as flush systems and toilets, require water, therefore.

The results of this study suggest that the age of household heads has significance in the household's sanitation facilities. The finding is consistent with studies conducted in Ethiopia [102] and Ghana [103], which suggest that the older the age of the household head, the more likely households are to have access to improved sanitation services. Some authors suggest that as people age, they make choices or adopt behaviors that improve their quality of life [104]. Another explanation would be that most elderly household heads belong to the active economic group and are therefore more inclined to install better sanitary facilities, particularly if they are homeowners [103]. Additionally, some young people have limited financial means, which may cause them to choose inexpensive homes, especially those that are rented, that could lack essential WASH facilities.

The feature importance analysis revealed that the mass media, such as radio and television, were another important predictor of sanitation facility utilization. This finding is supported by studies conducted in India [105] and Uganda [106], which indicate that mass media have a role in the household's sanitation utilization. Radio and television are powerful and efficient ways to conduct coordinated national awareness campaigns [107]. These devices can send messages to a huge number of people who are living in different locations at the same time. Radio is highly vital for reaching out with messages regarding sanitation facility access and health risks, especially in remote rural areas where there is limited electrical supply. Television is also very important for visualization or demonstration techniques, in addition to the messages.

Another significant importance was observed between hand washing facilities and household sanitation utilization. This finding is supported by a study conducted in Vietnam [108], Peru [109], and Ethiopia, which states the association between the presence of improved sanitation

facilities (latrine) in households and the availability of basic handwashing facilities. One possible explanation for this is that having a built-in latrine in a home encourages longer-term system use, which may make it easier to install permanent handwashing stations on the premises [107]. This specific finding suggests that efforts undertaken to expand latrine coverage worldwide would not only help to meet latrine coverage-driven targets but also increase basic handwashing facilities in the community. This particular finding implies that initiatives to enhance improved latrine coverage would contribute to the expansion of basic handwashing stations in the community as well as the achievement of latrine coverage-driven targets.

This study identified that marital status also influences a household's sanitation facility utilization. A study conducted in Bangladesh [110] suggested that the use of improved sanitation facilities is highly likely among married households as compared to single and divorced households. The possible explanation for this can be that in married households, there is a sharing of responsibility, knowledge, ideas, and financial burden, in which both partners will have some sort of income, which will help them to have financial stability, leading them to use improved sanitation facilities.

The other finding of our study indicates number of household member have a significant impact on the household's sanitation utilization. This finding was supported by the studies conducted in Tanzania [111] and Ethiopia [75], which also identified that family member has an impact on the household's sanitation facility use. When fewer family members share a latrine, the latrine is used less frequently overall, increasing the likelihood that it will be cleaner and potentially leading to better latrine utilization. However, a large family sharing a latrine increases the frequency of daily usage, which compromises an individual's sense of duty to use the latrine appropriately, increasing the likelihood that the latrine will be dirty. A study in the slums of Addis Ababa showed that a large family size of six or more persons led to poor hygiene of the latrines, which was in turn associated with diarrheal diseases [112].

This study reveals that the number of children under 5 also has a measurable impact on the household's sanitation facility utilization. This study is supported by a study conducted in Tanzania [111], Ethiopia, India, and Zambia together [113], and Ethiopia [114, 115], which also emphasizes that having under 5 children influences the household's sanitation facility utilization. Previous studies have also demonstrated that some parents' perception of a slight health danger from a child's feces, as well as their worries that their children may get the flu from the foul-smelling restrooms or fall into the pit, they let their young children practice open defecation [114].

While our study analyzed pooled DHS data across sub-Saharan Africa to identify regional-level determinants of sanitation facility access, we acknowledge that country-specific heterogeneity and spatial variations exist due to differences in governance, infrastructure, culture, and socioeconomic conditions. Our focus was on identifying overarching predictors that are broadly applicable across the region using by machine learning framework. Future research could build on our findings by integrating country-specific or spatially explicit models, such as geographically weighted regression or hotspot analysis, to explore intra-country inequities and inform targeted intervention.

Conclusion

This study utilized machine learning techniques to analyze the determinants of sanitation facility access in sub-Saharan Africa using recent DHS data, revealing that 49.85% of households use unimproved sanitation facilities. The Random Forest model was the most robust predictor (accuracy: 80.61%, F1-score: 0.8377), and SHAP analysis identified shared toilet use (feature importance=0.233), education level (0.204), and wealth index (0.094) as the strongest predictors, followed by residence, electricity access, age of the household head, drinking water source, household water treatment, and cooking fuel choice. Media access, sex of household head, hand-washing facility, water access, soap presence, location of water source, marital status, and number of household members also had notable influence, while the number of under-5 children showed minimal predictive impact.

Based on these findings, actionable strategies to reduce the use of unimproved sanitation facilities include prioritizing economic empowerment programs, such as cash transfers or livelihood support, to improve household affordability for improved latrines; implementing education and awareness campaigns through formal schooling, Community-Led Total Sanitation programs, and rural community outreach to encourage behavioral change; investing in rural water and sanitation infrastructure, including expanding piped water systems and subsidizing latrine construction, to bridge urban-rural disparities; and scaling up media-based interventions through radio, television, and social media to disseminate culturally tailored sanitation messages to households with media access. Targeted interventions should focus on households with shared toilets, low education, limited electricity access, or other high-risk characteristics identified in this study. By aligning policy and programmatic efforts with the most influential determinants identified, these evidence-based strategies can accelerate progress toward universal access to improved sanitation across sub-Saharan Africa.

Strengths and limitations of the study

This study has several merits. Firstly, to our knowledge, this is the first attempt to predict household sanitation facility access in sub-Saharan Africa using machine learning approaches, identifying determinant factors at a large regional scale. Therefore, it provides a valuable baseline for developing targeted interventions and tracking progress toward improved sanitation utilization. Secondly, the analysis utilized a large, nationally representative, weighted dataset (DHS), enhancing the generalizability and reliability of findings. Thirdly, the use of methodologically rigorous and advanced machine learning techniques achieved high predictive accuracy and identified key determinants of sanitation facility utilization. The application of SHAP values enhanced interpretability, highlighting socioeconomic, demographic, and environmental disparities.

However, several limitations should be noted. The study relied solely on DHS variables, which primarily capture socioeconomic and demographic characteristics and may not fully reflect contextual, cultural, or behavioral factors (e.g., local sanitation norms, community practices, or governance influences) that affect sanitation behaviors. This may introduce some bias and limit the model's capacity to capture local-level heterogeneity. Additionally, the cross-sectional design precludes causal inference, and although SMOTE was used to address class imbalance, synthetic samples may not entirely represent real-world variability. Future research integrating contextual and qualitative data could provide a more holistic understanding of sanitation facility utilization.

Abbreviations

ANN	Artificial neural network
DHS	Demographic and Health Survey
DT	Decision Tree
KNN	K-nearest neighbor
LR	Logistic regression
ML	Machine learning
MAPE	Mean absolute percent error
RF	Random forest
ROCAUC	ROC curve
SDG	Sustainable Development Goal
SHAP	Shapley Additive exPlanations
SMOTE	Systematic Minority Over-Sampling Technique
SSA	Sub-Saharan Africa
STATA	Statistical software for data science
XGBoost	Extreme gradient boosting

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Authors' contributions

G.Y. participated in conceptualization, data abstraction, selection, data extraction, data analysis, and manuscript writing. L.D., E.A.A., E.A.W., and M.T. participated in data analysis and manuscript writing. G.Y. guided the work and manuscript writing. All of the authors have read and approved the final manuscript.

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Data availability

The datasets used in this study are available from the DHS website [**<https://dhsprogram.com/Data/>**](<https://dhsprogram.com/Data/>) upon request from the DHS program team. Written permission to use the data was obtained from Measure DHS. All data generated or analyzed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

The study was made possible by obtaining a permission letter from www.dhsprogram.com after an online request for access to sub-Saharan African DHS data. The datasets were treated with total confidence. This study was based on secondary data from the sub-Saharan DHS. Informed consent, confidentiality, anonymity, and privacy of study participants are already ethically handled by the DHS office. Outside of this study, the data were not used or changed. Neither patients nor members of the public participated in this study.

Consent for publication

Informed consent and anonymity do not apply to our study, as these were obtained after the responsible implementing institution collected the data.

Competing interests

The authors declare no competing interests.

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